Eidolon: Topology-Aware Flexible Paxos for High-Latency, Intermittently Connected Systems

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Abstract

We study Paxos in a high-latency, intermittently disconnected setting motivated by Earth–Mars operations. The main claim is structural: safety depends on quorum intersection, not on a flat majority rule, so quorum shape can be aligned with physical topology. We evaluate Eidolon, a discrete-event simulator with a tiered quorum family in which the hot Phase 2 path stays entirely on Earth while the rarer Phase 1 path spans Earth, LEO, Moon, and Mars. The experiments compare hard conjunction blackouts with a repeater-assisted degraded-connectivity model. Across a 50-seed sweep, Earth-local latency remains stable at 251.0–251.0 ms mean and 263.6–263.8 ms p95 despite large Mars-delay changes. Hard blackouts drive during-blackout global success to 0% in every tested setting. Under repeater-assisted connectivity, during-blackout global success remains 100% for most tested points and degrades only at the most extreme cases (85.5%±3.5% at 750 s one-way Mars delay with an 1800 s blackout, and 82.5%±3.2% at 1342 s with an 1800 s blackout). These results position topology-aware Flexible Paxos as a useful design pattern for high-latency, intermittently connected deployments rather than a Mars-specific curiosity.

1 Introduction

Consensus in interplanetary settings is constrained less by logic than by geometry. If every quorum is treated as a flat majority over all nodes, then the steady-state commit path inherits the full cost of Earth–Mars propagation. Our thesis is that Paxos safety is compatible with a much more topology-aware arrangement: the hot commit path can remain local to the fast tier as long as Phase 1 and Phase 2 quorums still intersect.

The evaluation is implemented in Eidolon, the simulator used throughout this paper. The name is deliberate: during blackout and long-delay operation, every tier reasons over an inevitably stale projection of remote state. The central claim of the paper is that this is acceptable for safety if quorum geometry is chosen correctly.

This paper makes four concrete contributions. First, it gives an explicit quorum-family definition for the tiered construction used in the simulator. Second, it states the safety and liveness assumptions under which the construction is evaluated. Third, it reports a full parameter sweep over Mars latency, blackout duration, and timeout settings rather than a single representative run, using 50 seeds per point for the main experiments. Fourth, it ties each empirical claim to an executable artifact and output file.

2 Formal Model

2.1 System Model

Let the acceptor universe be

$$N = E \cup L \cup U \cup M,$$

where the tiers are pairwise disjoint and represent Earth, LEO, Moon, and Mars respectively. In the evaluated topology,

$$|E| = 5, \quad |L| = 1, \quad |U| = 1, \quad |M| = 3, \quad |N| = 10.$$

Earth is the fastest tier. The network is asynchronous: messages experience delay and jitter, and links may be removed during scheduled blackout windows. Acceptors are crash-stop and non-Byzantine, and accepted Paxos state is stable once written. Timeouts affect liveness only; they do not affect safety.

The simulator uses three logical consensus scopes.

- Earth-local consensus runs over the Earth tier with Flexible Paxos, using $q_1 = 4$ and $q_2 = 2$ over the five Earth acceptors.
- Mars-local consensus runs over the three Mars nodes with a majority quorum of size two.
- Global reconciliation runs over all ten nodes with the topology-aware quorum family defined below.

2.2 Global Quorum Families

Let the Phase 2 quorum family be

$$\mathcal{Q}_2 = \{E\},$$

that is, the full Earth tier. This keeps the hot commit path entirely on the fast tier.

The structural Phase 1 family is

$$\widehat{\mathcal{Q}}_1 = \{Q \subseteq N \mid Q \cap E \neq \emptyset,\ Q \cap L \neq \emptyset,\ Q \cap U \neq \emptyset,\ Q \cap M \neq \emptyset\}.$$

Every Phase 1 quorum spans all tiers. The implementation uses a conservative strengthening of this family:

$$\mathcal{Q}_1 = \{Q \in \widehat{\mathcal{Q}}_1 \mid |Q| \geq |N| - |E| + 1 = 6\}.$$

Operationally, this means "one node from every tier plus enough extra Earth nodes to satisfy the standard sufficient size condition." In the evaluated 3/1/1/5 topology, the minimum implemented Phase 1 quorum contains one Mars node, the Moon node, the LEO node, and three Earth nodes.

2.3 Safety Argument

Proposition 1. Every global Phase 1 quorum intersects every global Phase 2 quorum.

Proof sketch. Let $Q_1 \in \widehat{\mathcal{Q}}_1$ and $Q_2 \in \mathcal{Q}_2$. By definition, Q_1 contains at least one Earth node and $Q_2 = E$, so $Q_1 \cap Q_2 \neq \emptyset$. Because $\mathcal{Q}_1 \subseteq \widehat{\mathcal{Q}}_1$, the same result holds for the implemented family. Therefore the cross-phase intersection requirement used by Flexible Paxos is preserved. □

This is the core modeling point: majority size is only one sufficient route to intersection. Here the relevant invariant is structural overlap between the tier-spanning Phase 1 family and the Earth-only Phase 2 family.

Table 1: Experimental design.

Item	Sweep study	Liveness-envelope study
Purpose	Compare hard blackout vs repeater over topology and blackout severity	Identify timeout thresholds for during-blackout success
Scenarios	`blackout_only`, `with_repeater`	`blackout_only`, `with_repeater`
Varied parameters	Mars one-way latency $\in \{186, 750, 1342\}$ s; blackout duration $\in \{300, 900, 1800\}$ s	Global timeout $\in \{120, 240, 360, 500, 720\}$ s; blackout duration $\in \{300, 900, 1800\}$ s
Fixed parameters	Blackout start 600 s; simulation end 4000 s; reconcile interval 120 s; global max rounds 1	Mars one-way latency 186 s; blackout start 600 s; simulation end 4000 s; reconcile interval 120 s; global max rounds 1
Seeds	40–89 (50 runs per point)	40–89 (50 runs per point)
Topology	5 Earth, 1 LEO, 1 Moon, 3 Mars nodes	Same as sweep study
Primary metrics	Earth/Mars local success, global pre/during/post success, recovery lag, average global latency, local average and p95 latency	During-blackout and post-blackout success, recovery lag, average global latency
Runtime control	One CSV row per seed and scenario, plus aggregated mean/95% CI CSV	One CSV row per seed and scenario, plus aggregated mean/95% CI CSV

2.4 Liveness Assumptions

The evaluation does not prove unconditional liveness. Instead, it studies liveness envelopes under explicit assumptions.

- Earth-local progress requires enough reachable Earth acceptors to form the Earth-local Flexible Paxos quorums.
- Mars-local progress requires a reachable Mars majority.
- Global progress requires a successful Phase 1 exchange across all tiers and a timeout budget that exceeds the realized end-to-end delay on the currently available path.
- In the `blackout_only` scenario, Earth–Mars communication is removed during conjunction.
- In the `with_repeater` scenario, blackout is replaced with a degraded Earth–relay–Mars path.

3 Experimental Design

We use the repository's discrete-event simulator and the exact commands recorded in `docs/step9-repro.md`. Table 1 separates fixed parameters from the parameters varied in each experiment family.

The repeater model is an explicit alternate path in the simulator, not a scalar multiplier on the direct Earth–Mars links. In the `with_repeater` scenario, Earth reaches a `lagrange-relay` hop in 350–360 ms one way, and the relay reaches Mars in 220 s one way with 10 s jitter. In the `blackout_only` scenario, that path is absent and the relevant cross-tier links are removed during the conjunction window. The relay is therefore modeled as a separate degraded control-plane path with its own latency distribution.

Table 2: Local-path stability across all 18 sweep points (means and p95 values aggregated from `step9_sweep_ci.csv`, 50 seeds per point).

Metric	Minimum	Maximum
Earth local mean latency	251.0 ms	251.0 ms
Earth local p95 latency	263.6 ms	263.8 ms
Mars local mean latency	21.1 ms	21.1 ms
Mars local p95 latency	22.6 ms	22.7 ms
Earth local success rate	100%	100%
Mars local success rate	100%	100%

The sweep study writes raw and aggregated `step9_sweep` CSVs under `results/step9/`. The timeout study writes raw and aggregated `step9_liveness` CSVs in the same directory. Plot files are generated under `results/step9/plots/`.

4 Results

4.1 Local Autonomy Is Stable

Table 2 summarizes the local-path metrics across the full 18-point sweep. Earth-local latency is essentially invariant to Mars delay and blackout duration, which is the expected consequence of keeping the global Phase 2 path on Earth. Mars-local metrics are also stable because Mars-local consensus uses its own local majority.

4.2 Hard Blackouts Remove Global During-Blackout Progress

Table 3 reports the full hard-blackout sweep. During-blackout global success is 0% in every tested condition. At the closest Mars-delay setting (186 s one-way), the system resumes post-blackout reconciliation, but recovery lag grows as blackout duration lengthens. At the 750 s and 1342 s settings, the 500 s global timeout budget is insufficient even after blackout, so no global success is observed.

4.3 Repeater-Assisted Connectivity Preserves Most of the Liveness Envelope

Table 4 reports the repeater-assisted sweep. The qualitative pattern is different: during-blackout success is 100% for most parameter points, including the 750 s and 1342 s latency cases at 300 s and 900 s blackout durations. Degradation appears only at the most extreme 1800 s blackout settings. Post-blackout success is still limited by the fixed 500 s global timeout budget at the larger Mars-delay points.

4.4 Timeout Thresholds Show a Sharp Liveness Transition

Figures 1 and 2 visualize the timeout sweep at 186 s one-way Mars delay. Both scenarios fail during blackout for timeout budgets of 120–360 s. The threshold appears between 360 s and 500 s. At 500–720 s, hard blackout still yields no during-blackout success, but the repeater-assisted model succeeds during the 900 s and 1800 s windows.

The 300 s blackout line in the timeout sweep deserves care: the aggregated CSV contains zero during-blackout attempts at that setting, so its reported 0% should be read as "no attempts completed during the window" rather than "attempts failed despite relay connectivity." This is

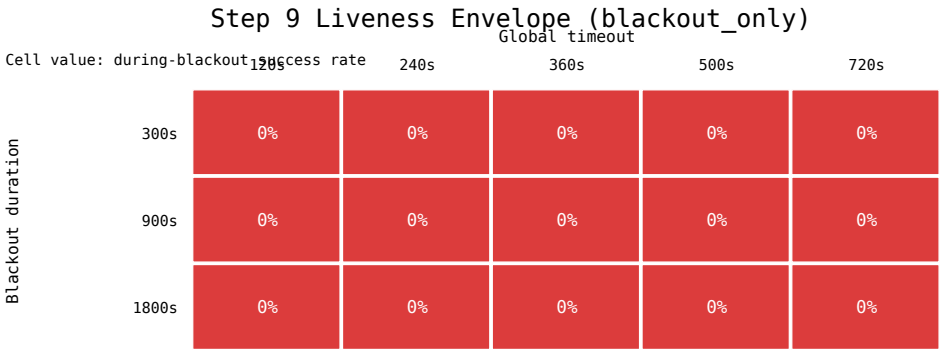

Figure 1: During-blackout global success under the hard-blackout model.

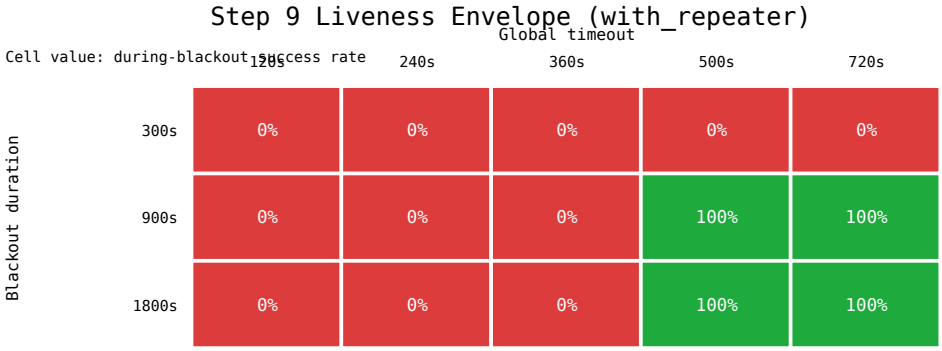

Figure 2: During-blackout global success under the repeater-assisted model.

a workload-scheduling artifact of the 120 s reconciliation interval combined with a short blackout window and belongs in the interpretation of the figure rather than the safety claim.

The 750 s and 1342 s rows in Tables 3 and 4 expose a second, more substantive effect: with a fixed 500 s global timeout, the system can fail to recover global reconciliation even after blackout ends. This is not a blackout-only phenomenon; it is a timeout-mismatch phenomenon. Operationally, that suggests at least three follow-on strategies: increase the timeout budget with distance, adapt timeouts dynamically from observed round-trip behavior, or defer global reconciliation attempts entirely when the system is known to be outside the current timeout envelope.

5 Related Work

Paxos established the standard safety argument for replicated state machines under asynchronous messaging and crash failures [10]. Flexible Paxos relaxed the "every quorum is a majority" intuition by showing that safety only requires Phase 1/Phase 2 cross-intersection, not universal pairwise intersection among all quorums [8]. Our construction follows that line directly: the hot path is

made small and local, while the slower election path is made larger and topology-spanning.

The broader quorum-systems literature provides the combinatorial foundation for moving beyond majority quorums. Grid-style and crumbling-wall constructions show that quorum geometry can be engineered around availability and structure rather than simple counts [3, 13]. What this paper adds is a concrete mapping from that literature to a tiered Earth/LEO/Moon/Mars deployment, together with an explicit distinction between the conceptual structural family and the more conservative implemented family used in the simulator.

Geo-distributed consensus systems provide the closest practical analog. WPaxos [1] uses Flexible Paxos with object-stealing to keep the commit path in the nearest region. EPaxos [12] removes the leader bottleneck so that any replica can commit locally when commands do not conflict. Mencius [11] distributes the leader role round-robin across WAN sites. MDCC [9] achieves single-round cross-datacenter commits through Generalized Paxos. All four systems optimize for heterogeneous WAN latency in the tens-to-hundreds of milliseconds range and assume that all replicas are reachable most of the time. Eidolon differs in two ways. First, the asymmetry is far more extreme: the one-way delay spans three orders of magnitude (sub-second Earth to 1342 s Mars opposition) rather than the 2–5× variation across datacenters. Second, the target phenomenon is scheduled total disconnection—conjunction blackouts lasting minutes to hours—rather than merely heterogeneous latency. Even so, the structural question is the same: which coordination steps belong on the hot path, and which can be pushed onto a slower or rarer control path?

Delay-tolerant networking motivates the operating regime but addresses a different layer of the problem. The DTN architecture [2, 6] studies intermittent connectivity, long propagation delay, and store-and-forward delivery across challenged links. The overwhelming majority of DTN work concerns routing and custody transfer, not consensus over shared state. Where consistency has been studied in DTN-like settings, the dominant approach is to abandon consensus entirely in favor of conflict-free replicated data types (CRDTs) that guarantee eventual convergence through algebraic merge properties [7]. SwarmDAG [15] takes a middle path: a partition-tolerant distributed ledger for robot swarms that uses extended virtual synchrony to handle split-and-merge dynamics, with Tran et al. separately measuring consensus latency under network partitioning [14]. Protocol Labs' Interplanetary Consensus (IPC) [5] uses hierarchical subnets with independent consensus for blockchain scaling; despite the name, it does not target interplanetary latencies. NASA's DSN replication study [4] evaluates commercial database replication for ground-station coordination but does not formalize quorum geometry. Our contribution occupies the gap between these approaches: we show that Paxos safety is compatible with topology-aware quorum geometry under scheduled disconnection, without surrendering to eventual consistency or restricting the protocol to a single tier.

6 Threats to Validity and Model Limitations

The results are design-level, not deployment-level. Several limitations matter.

First, the network model is deliberately abstract. Blackout is represented as deterministic link removal, and repeater assistance is represented as a fixed degraded path. Orbital dynamics, antenna scheduling, contention, coding strategy, and variable link availability are not modeled. A more realistic space-network study could change the exact liveness thresholds even if the quorum-intersection argument stays intact.

Second, the workload is stylized. Earth and Mars local proposals run on fixed cadences, and global reconciliation runs on a fixed 120 s interval with bounded retries. This creates aliasing effects, most visibly in the 300 s blackout cases of the timeout sweep, where the measured during-

blackout rate is driven by whether an attempt is scheduled entirely inside the window. That effect is informative about operational tuning, but it should not be over-interpreted as a sharp physical threshold.

Third, the study fixes one topology size and one global Phase 2 family. We do not sweep Earth-tier size, Mars-tier replication degree, or alternative relay placements. The local autonomy result is therefore strongest as a claim about this class of construction, not as a proof that the chosen 5/1/1/3 split is optimal.

Fourth, the fault model is crash-stop only. Byzantine behavior, correlated software faults, reconfiguration during blackout, and storage failures are all outside scope. The paper therefore speaks to quorum geometry under delay and partition, not to adversarial resilience.

Finally, some metrics remain higher-variance than others even after increasing the main experiments to 50 seeds per point. The near-deterministic local-path metrics are now tightly estimated, and the noisier recovery metrics are substantially better resolved than in the initial 5-seed pass. For example, the repeater-assisted 186 s/300 s recovery lag is now 38.3 s with a 95% confidence interval of ± 31.8 s. That is much more interpretable than the initial wide interval, but it still indicates that recovery behavior is meaningfully more variable than the steady-state local path. We therefore report means with 95% confidence intervals throughout and treat recovery estimates as empirical envelope measurements rather than exact deployment predictions.

An additional modeling tradeoff is the strict Phase 2 choice $\mathcal{Q}_2 = \{E\}$. Using the full Earth tier as the only Phase 2 quorum makes the hot path clean and easy to analyze, but it also means any missing Earth acceptor blocks global Phase 2 progress. A relaxed Earth-local family such as 4-of-5 Earth nodes could still preserve cross-phase intersection with an appropriately chosen Phase 1 family. We do not explore that availability/latency tradeoff in the current paper.

7 Discussion: Eventual Consistency Under Topology

One useful way to read the evaluated construction is as a seed for analyzing eventual consistency in topologically asymmetric systems. During conjunction blackout, Earth and Mars are not merely "temporarily inconsistent replicas" in the usual flat sense. Each tier remains internally strongly consistent according to its local quorum policy, while cross-tier state becomes stale in a structured way determined by the unavailable global reconciliation path. That suggests a topology-scoped view of consistency: quorum geometry determines which consistency domains remain strong during disruption, and the network topology determines how and when those domains can be stitched back together.

The present simulator is not rich enough to make strong claims about application-level merge semantics, conflict resolution, or arbitrary eventual-consistency protocols. It does, however, suggest a useful analysis direction. Standard eventual-consistency descriptions often treat the inconsistency window as a temporal black box. In contrast, the results here make visible the internal structure of that window: local strong-consistency islands, explicit stale views of remote state, and a measurable protocol-level reconciliation cost once cross-tier communication resumes. We therefore view the current paper not as a contribution to eventual-consistency theory per se, but as evidence that interesting future analyses should characterize consistency locality and reconciliation behavior as functions of topology rather than as purely global or time-only properties.

8 Conclusion

The evaluated construction shows that high-latency consensus should be framed as a quorum-shaping problem, not as a flat-majority problem. By placing the global Phase 2 quorum entirely on Earth and making Phase 1 tier-spanning, the design preserves Paxos safety while keeping Earth-local behavior nearly constant across large Mars-delay changes. Hard blackouts eliminate global during-blackout progress; repeater-assisted degraded connectivity restores most of that liveness envelope, though only inside the configured timeout regime. The broader lesson is that topology-aware Flexible Paxos offers a principled way to separate hot-path latency from rare cross-tier coordination in high-latency, intermittently connected systems.

A Claim-to-Artifact Traceability

Table 5 maps the paper's main claims to scripts and output files in the repository. Artifact paths are relative to the repository root.

Table 5: Claim-to-artifact traceability.

Claim	Evidence in repository	Primary artifact(s)
Earth-local latency stays near 251 ms mean and 264 ms p95 across Mars-delay changes.	Aggregated sweep rows for all 18 scenario/latency/blackout combinations; columns `earth_local_avg_latency_s_mean` and `earth_local_p95_latency_s_mean`.	`results/step9/step9_sweep_ci.csv`; generated by `experiments/step9_sweep.py`.
Hard blackout produces zero during-blackout global success throughout the sweep.	All `blackout_only` rows have `global_during_rate_mean = 0`.	`step9_sweep_ci.csv`.
Repeater-assisted connectivity preserves during-blackout success for most tested settings and degrades only at the 1800 s extreme cases.	`with_repeater` rows show 100% during-blackout success except 85.5%±3.5% at 750/1800 and 82.5%±3.2% at 1342/1800.	`step9_sweep_ci.csv`; plotted by `experiments/plot_step9.py`.
Timeout threshold appears between 360 s and 500 s at the 186 s Mars-delay setting.	Liveness-envelope rows show zero during-blackout success at 120–360 s and non-zero repeater-assisted success at 500–720 s for the longer blackout windows.	`step9_liveness_ci.csv`; plots `liveness_envelope_*.svg`.
The paper is reproducible from repository commands.	Exact commands for environment setup, sweep execution, and plot generation.	`docs/step9-repro.md`.

Claim	Evidence in repository	Primary artifact(s)
The implemented global quorum family is topology-aware rather than size-only.	Global proposer instantiates `CrumblingWallQuorum`; quorum validation uses `is_phase1_quorum` and `is_phase2_quorum`.	`demo_step_9.py`; `quorums.py`.

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Table 3: Hard-blackout sweep (`blackout_only`, mean±95% CI over 50 seeds).

Mars delay (s)	Blackout (s)	During-blackout success	Post-blackout recovery lag (s)
186	300	0%±0.0%	577.7±1.0
186	900	0%±0.0%	597.7±1.1
186	1800	0%±0.0%	937.7±0.9
750	300	0%±0.0%	n/a
750	900	0%±0.0%	n/a
750	1800	0%±0.0%	n/a
1342	300	0%±0.0%	n/a
1342	900	0%±0.0%	n/a
1342	1800	0%±0.0%	n/a

Table 4: Repeater-assisted sweep (`with_repeater`, mean±95% CI over 50 seeds).

Mars delay (s)	Blackout (s)	During success	Post success	Recovery lag (s)
186	300	0%±0.0%	100%±0.0%	38.3±31.8
186	900	100%±0.0%	100%±0.0%	264.4±66.2
186	1800	100%±0.0%	100%±0.0%	232.7±3.4
750	300	100%±0.0%	0%±0.0%	192.2±2.1
750	900	100%±0.0%	0%±0.0%	185.2±3.5
750	1800	85.5%±3.5%	0%±0.0%	n/a
1342	300	100%±0.0%	0%±0.0%	193.1±1.9
1342	900	100%±0.0%	0%±0.0%	183.3±2.7
1342	1800	82.5%±3.2%	0%±0.0%	n/a